A bipolar charge architecture in the aRES N-terminal block

Rotation project `aRES_ppi`

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Sequence composition and AlphaFold3 model analysis only — no experimental data. *E. coli* UC5 aRES, 459 aa.

1. The N-terminal block is two modules of opposite charge

Per-region charge composition, counting Lys+Arg against Asp+Glu:

Region	Length	K	R	D	E	Net (per 100 aa)
Zn ribbon 1–48	48	10	3	2	0	+11 (+22.9)
NTD-1 49–120	72	0	3	11	8	−16 (−22.2)
N-block 1–120	120	10	6	13	8	−5 (−4.2)
RES side 126–447	322	23	19	24	21	−3 (−0.9)

Two directly adjacent modules carry almost exactly equal and opposite charge density. The zinc ribbon has ten lysines and *no glutamate*; NTD-1 has nineteen acidic residues and *no lysine at all.* The RES domain next door is at −0.9 per 100 residues, i.e. an ordinary protein surface — so the extremity is confined to the N-block and is anticorrelated between its halves.

Residues 31–40 are `CSYC KRKRK N`: a five-residue basic run immediately C-terminal to the second CxxC motif. That is the canonical arrangement of a nucleic-acid-binding zinc ribbon, and it is a stronger signal than the net charge.

2. Both charged faces are solvent-exposed

In the electrostatic surface of the monomer model, the basic ribbon face (carrying the zinc) and the acidic NTD-1 face sit on adjacent lobes and each faces solvent. They do not neutralise one another in an intramolecular clamp. The N-block therefore presents a **bipolar surface**: one strongly basic face and one strongly acidic face, both available for binding.

3. Working model — hypothesis, not a result

The N-block may recognise **a polymerase engaged with nucleic acid**, with the two modules reading the two components:

- the **basic zinc ribbon** binds the DNA;
- the **acidic NTD-1** binds the polymerase's basic nucleic-acid-binding cleft, acting as a DNA mimic (the logic of Ocr, T7's own gene 0.3 product, which mimics B-form DNA to inhibit host EcoKI; protein mimics of DNA are acidic).

What recommends it:

- **Promiscuity.** Every polymerase has a basic DNA-binding cleft regardless of sequence family. aRES must fire on T7 gp5, on divergent T-even and T5 polymerases, and on something in λ. Recognising a conserved *physical* feature does not require a conserved epitope.
- **Specificity.** Requiring both faces to engage is an AND-gate — which is what the system needs, since the host cytoplasm is full of polymerases and a false positive is suicide.
- **Why the sensor is bimodular at all**, which neither a folding-scaffold nor an effector role for the ribbon explains.
- **λ.** Host Pol III hijacked at oriλ by P/O is still a polymerase on DNA in an aberrant context, consistent with the host-factor-hijacking model.
- **It survives the DNA-free reconstitution.** Purified gp5·TrxA activates aRES with no DNA and no dNTPs. If NTD-1 is the DNA mimic it engages the cleft without DNA present; the ribbon's DNA arm would contribute avidity and context-specificity *in vivo*. A sufficient minimal reaction and the full mechanism need not be the same thing.

4. Tests, cheapest first

1. **Artifact control on the electrostatics.** Recompute with the Zn deleted, and colour the surface by residue type as a model-free check (`analysis/scripts/color_by_restype.pml`). The blue is currently centred on the metal, and if the four Cys were not treated as thiolates the site is over-positive by up to +6.
2. **Back face and contiguity.** If the two patches form one contiguous bipolar surface, the two-sided model is viable; if they point away from each other they cannot engage one target at once and the model needs rethinking.
3. **Is D104 on the acidic face?** The existing cofold places aRES D104 against gp5 S122. If D104 sits in the exposed acidic patch, the model connects to the one interface result in hand, and the D104A/D104N experiment tests the model rather than a single weak contact.
4. **AF3 cofold of aRES with a primer–template duplex.** Direct test: does DNA dock onto the KRKRK face? One server job.
5. **DNA binding, 1–120 vs 49–136.** EMSA or fluorescence anisotropy against ssDNA, dsDNA and a primer–template fork. 49–136 contains no ribbon and no zinc, so it is the built-in negative control and is already in the construct series.
6. **DNA competition of the aRES–gp5 interaction.** If NTD-1 occupies the DNA cleft, adding primer–template should compete aRES off gp5. Sharpest available test of the mimicry half.
7. **Salt dependence of K_d.** An electrostatically driven interface is strongly ionic-strength sensitive; a conventional epitope interface much less so.
8. **Conservation.** Compute the same per-module charge split across PF18870 homologs. If ribbon-containing homologs consistently show the mirror, it is functional architecture rather than an accident of one sequence.

5. Caveats

- This is compositional reasoning over a single predicted model. It generates a hypothesis; it does not confirm one.

- The electrostatic rendering may overstate the basic patch at the zinc (test 1).
- The surface was inspected in one orientation (test 2).
- gp5 D106 is itself acidic, which is awkward for a naively electrostatic interface. Local context will matter more than net charge.
- The DNA-free *in vitro* activation bounds any model in which nucleic acid is the trigger for T7. It does not exclude a contribution to affinity or to other triggers.

6. Figures to make

1. `figs/restype_front.png`, `figs/restype_back.png` — surface coloured by residue type, front and back of the N-block. Script: `analysis/scripts/color_by_restype.pml`.
2. `figs/apbs_nozn.png` — the existing electrostatic surface recomputed with the Zn removed, same orientation and same $\pm 5\, k_{\mathrm{B}}T/e$ scale, beside the original.
3. `figs/charge_window.png` — net charge in a 15-residue sliding window along residues 1–459, with the module boundaries marked. Shows the anticorrelation at 48 as one line.
4. `figs/nblock_cartoon.png` — cartoon of 1–120 with the ribbon, NTD-1 and the zinc coloured, oriented to match figure 1.